General Mathematics — Exercise Sheet 22

Adjugates and equivalent descriptions

Variant: Statements

Exercise 1 (A column with two nonzero entries). For the real matrix

$$D = \begin{pmatrix} 2 & 0 & 1 & 0 \\ 1 & 3 & 0 & 2 \\ 0 & 0 & 4 & 0 \\ 2 & 1 & 1 & -1 \end{pmatrix},$$

compute the cofactors C_{24} and C_{44} and use expansion along column 4 to find $\det D$. Display the deleted submatrices and keep their order unchanged. State whether $Dx = b$ has one solution for every $b \in \mathbb{R}^4$.

Exercise 2 (A singular adjugate can supply a direction). For $t \in \mathbb{R}$, consider

$$B_t = \begin{pmatrix} t & 1 \\ t^2 & t \end{pmatrix}.$$

Compute $\det B_t$ and $\operatorname{adj} B_t$. Verify both adjugate product identities directly. Show that the second column of the adjugate is nonzero and forms a one-column basis of $\ker B_t$ for every t, including $t = 0$. Explain why the inverse formula is unavailable despite the nonzero adjugate.

Exercise 3 (Cramer's rule with a right-side parameter). For every $a \in \mathbb{R}$, solve

$$x_1 + x_2 = a, \qquad 2x_2 + x_3 = 1, \qquad 2x_1 + x_3 = 2$$

by Cramer's rule. Display all three replacement-column determinants. Find exactly the values of a for which all three solution coordinates are nonnegative, including any boundary values.

Exercise 4 (Adjugate and adjoint are different). For the complex matrix

$$G = \begin{pmatrix} 1+\mathrm{i} & 1 \\ 0 & 2-\mathrm{i} \end{pmatrix},$$

compute $G^\dagger$, $\operatorname{adj} G$ and G^{-1}. Explain which operation uses complex conjugation and which uses cofactors. Use the adjugate inverse formula, and test whether $G^\dagger$ itself is an inverse by computing one suitable product entry. Use complex arithmetic and conjugation from Lecture 15.

Exercise 5 (The formal cross-product determinant and a basis). In $\mathbb{R}^3$, let e_1, e_2, e_3 be the standard coordinate columns and let

$$u = (2, -1, 1), \qquad v = (1, 3, -2), \qquad w = (3, 1, 0).$$

The *formal determinant* with first row e_1, e_2, e_3 means first-row expansion with those vector coefficients. It is a notation for a vector-valued expression, not a scalar determinant over $\mathbb{R}$. Use this expansion to compute $h = u \times v$, and compare the result with Lecture 13's component formula. Check $h \cdot u = h \cdot v = 0$. Find the signed and ordinary volume of the ordered columns u, v, w. Show that u, v, h form a basis of $\mathbb{R}^3$, and use Cramer's rule to find the coordinates of w in that ordered basis.